

SUNNY ZHOU

Machine Learning Engineer ◇ U.S. Citizen ◇ Berkeley, CA
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EDUCATION

Brandeis University, Michtom School of Computer Science
Master of Science in Computational Linguistics

Aug 2023 – May 2025

University of California, Davis, College of Letters and Sciences
B.S. in Mathematical and Scientific Computation; B.A. in Linguistics

Sep 2019 – Jun 2023

SKILLS

- **Programming Languages:** - Python - Java - C - R - MATLAB - HTML - TypeScript - CSS
- **Libraries** - numpy - pandas - scikit-learn - pytorch - tensorflow - transformers - langchain - vllm - Flask - FastAPI - streamlit - sqlalchemy - sqlite - tableau
- **Technical Skills** - Deep Learning - Model Training & Fine-Tuning (PEFT/LoRA) - Model Evaluation & Error Analysis - Dataset Construction & Curation - Prompt Engineering - RAG & Vector Embeddings - Statistical Analysis - Data Pipeline Development - System Design - Docker - API Development

PROJECTS

ZYC at SemEval-2026 Task 5: Application of BERT-based Contextual Embeddings Similarity for WSD 2026

- Designed and compared four novel transformer-based architectures for word sense disambiguation, exploring methods of capturing information with in narrative context.
- Improved model generalization by applying masked language modeling, gradual layer unfreezing, and targeted data augmentation, reducing overfitting on a small labeled dataset, achieving **57% accuracy** on test set.

Ambiguity and Bias: A Departure from Classical Evaluation – Master’s Thesis

2025

- Crafted and defined a novel dataset and evaluation metrics for a previously unmeasured task, probing semantic knowledge in **Llama-3.3**, **DeepSeek-R1**, **Gemma-3**, and **Mistral-3.1** — demonstrating experience designing benchmarks for novel ML problems.
- Ran systematic error analysis across all four models using **WordNet**-derived ground truth, surfacing consistent failure modes and quantifying performance gaps across word sense categories.

Production-Style RAG System for Code Repository Q&A

2025

- Built a full-stack retrieval system covering document ingestion, chunking, and **HNSW vector indexing** of source code and README files, enabling natural language queries over a codebase.
- Deployed a **Streamlit** chat interface backed by a quantized **Llama** model, taking the system from initial prototype to a fully functional, user-facing product.

Agentic LLM Tutoring System with Step-by-Step Reasoning

2025

- Shipped an end-to-end agentic tutoring system using **Llama-3.3** and a custom **LangChain** tool suite, with a **Flask** web interface serving MathJax-rendered step-by-step solutions.
- Increased solution accuracy through iterative prompt evaluation, identifying failure patterns and refining tool descriptions to prevent premature answer disclosure.

Novel Multilingual Dialogue Act Dataset from NPR Podcast Transcripts

2024

- Designed an annotation schema for a novel dialogue act classification task, scraping and processing multilingual NPR transcripts with **BeautifulSoup4** and authoring labeling guidelines used across multiple annotators.
- Validated dataset quality using **Fleiss Kappa** inter-annotator agreement scoring, running adjudication rounds to resolve label conflicts and produce a clean corpus for downstream training.

EXPERIENCE

Substitute Teacher Twin Rivers Unified School District, McClellan, CA

Apr 2026 – Present

- Delivered instruction across seven grade levels and special education in a high-need public school district serving a diverse, multilingual, and economically disadvantaged student population in northern Sacramento County.
- Adapted lesson plans and classroom management strategies across elementary, middle school, and special education settings on short notice.

Teaching Assistant Brandeis University, Waltham, MA

Aug 2023 – Dec 2024

Courses: NLP I, Numerical Methods and Big Data, Linear Algebra, Semantics

- Taught and explained deep learning, optimization, and NLP algorithms to graduate and undergraduate students across four courses, holding weekly office hours and recitations.
- Co-designed homework assignments, grading rubrics, and solution sets with faculty, contributing to curriculum development across ML and mathematics courses.

Undergraduate Research Assistant University of California, Davis, Davis, CA

Mar 2021 – Jun 2023

Language, Logic, and Learning Lab (L3Lab)

- Designed and ran psycholinguistics experiments on pragmatic inference in children, co-authoring study procedures and presenting findings at the UC Davis Language Symposium.